

TIC 66716482

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 8.028259 d
Epoch (t0) 2561.2964 BTJD
Duration 0.1200 d
Depth 3.329 % (33,287 ppm)
SNR 64.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

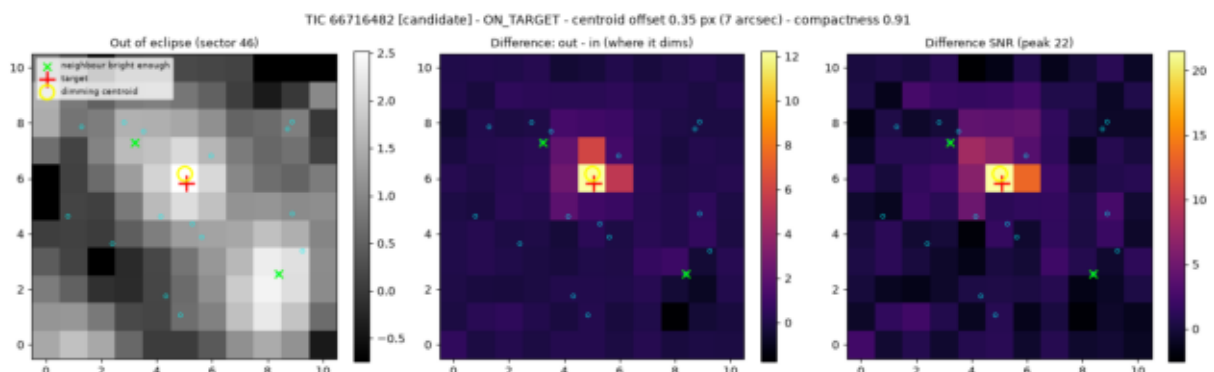
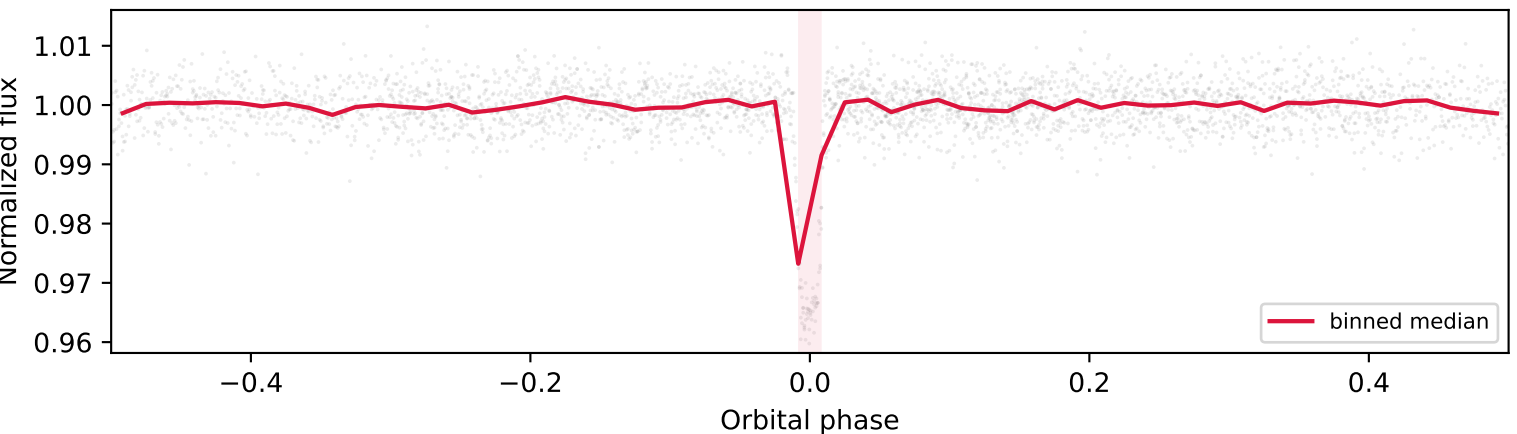
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 56.0 (gate ≥ 7.0)
Odd/even z-score 0.64 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 44.5 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 21.51 (gate ≥ 5.0) Compactness 0.91 (gate ≥ 0.50)
Dimming centroid offset 0.35 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 66715933
2.13 px away, dTmag +2.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.